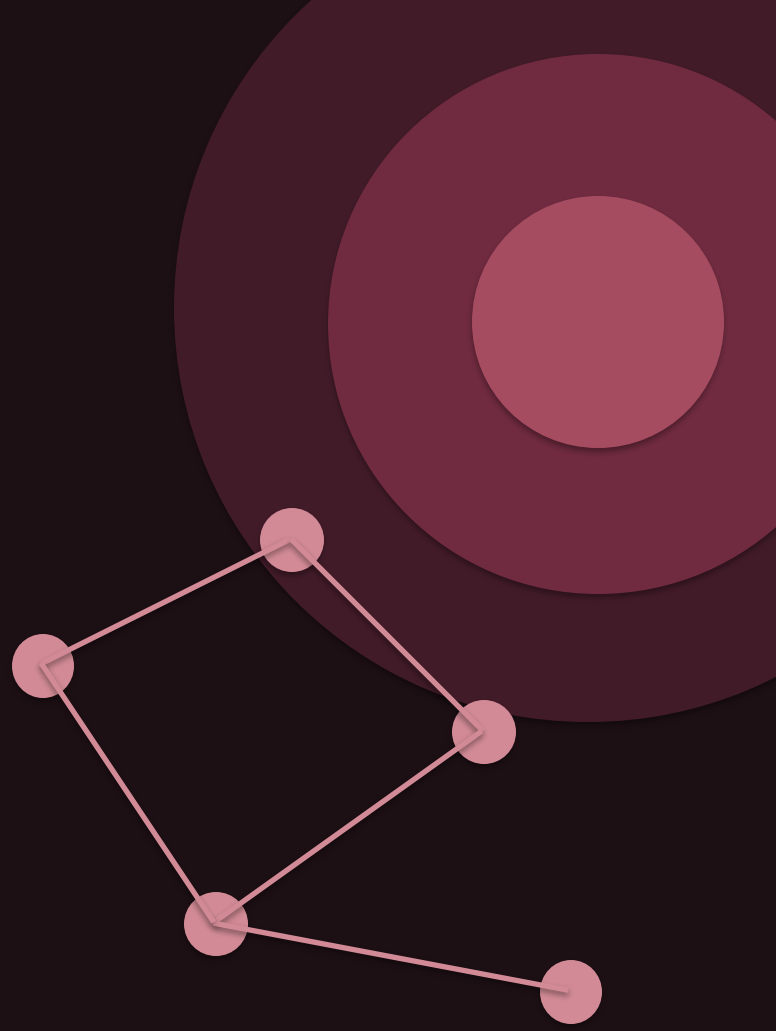
Supervised learning

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CLASS: BTECH(2ND YEAR) CST
TOPIC: FINANCIAL PREDICTION & FRAUD DETECTION

FINTECH • ARTIFICIAL INTELLIGENCE

AI-Powered Financial\nPrediction & Fraud Detection

A data-driven system for forecasting financial trends and identifying suspicious transactions in real time.



Concept • Architecture • Models • Real-world evidence • Deployment

One platform, two financial intelligence jobs

PREDICT

Estimate cash flow, revenue, demand, spending and risk before decisions are made.

DETECT

Score every transaction for unusual patterns and stop, verify or allow it.

LEARN

Use confirmed outcomes and analyst feedback to improve models continuously.

Expected business value

- Lower fraud losses and chargeback exposure
- Fewer false declines for genuine customers
- Better liquidity, budgeting and risk planning

CORE DESIGN PRINCIPLE

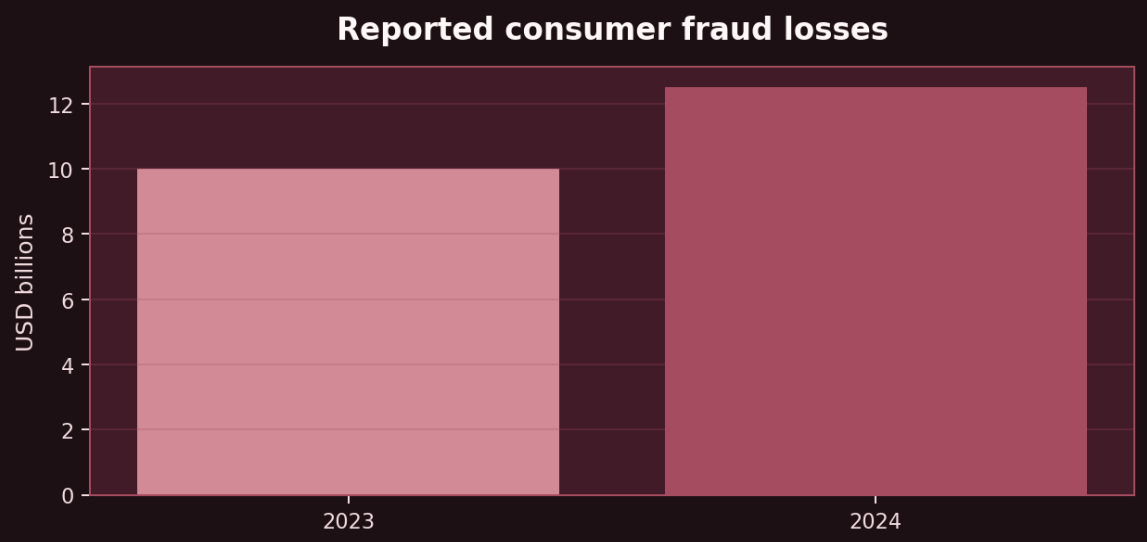
Do not treat a model score as a final verdict. Combine AI scores with business rules, customer verification and human review.

How financial fraud controls evolved

1950s–1980s	**Manual review & paper trails**	People inspected slips, signatures and account records. Slow and difficult to scale.
1980s–1990s	**Rule-based monitoring**	Velocity limits, country blocks and spending thresholds automated basic alerts.
1990s–2000s	**Chip cards & online commerce**	More digital payments increased speed and attack surfaces; EMV chip adoption strengthened card-present authentication.
2010s	**Machine learning at scale**	Models learned customer patterns from transaction histories; anomaly detection and ensemble models improved ranking.
2020s–today	**Real-time AI + graph signals**	Streaming systems, device intelligence, behavioural biometrics and network links help identify coordinated fraud.

Evolution is cumulative: modern systems still use rules, but add data, models and rapid feedback.

Why the problem is now critical



$12.5B

Reported US consumer fraud losses in 2024.

+25%

Increase in reported losses versus 2023.

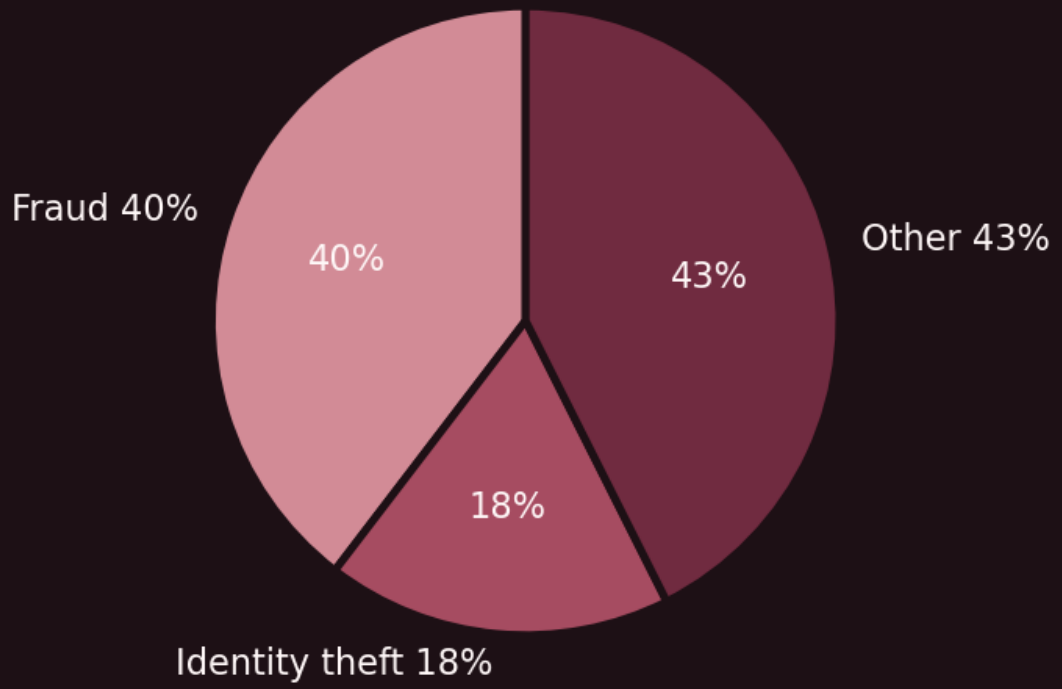
What makes the problem hard?

- Transactions happen in milliseconds.
- Fraud patterns change faster than static rules.
- Fraud is rare compared with legitimate activity.
- Blocking genuine customers also creates real cost.

Source: US Federal Trade Commission (FTC), 2024 Consumer Sentinel data; losses are reported by consumers, not a complete census.

The fraud landscape: multiple attack paths

FTC Consumer Sentinel reports, 2024

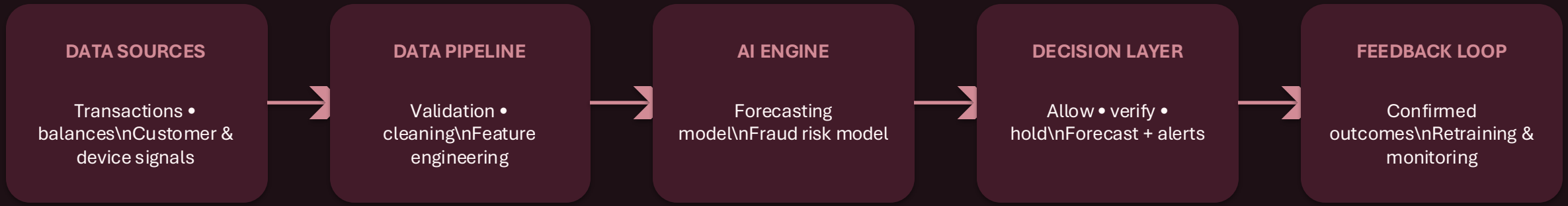


FTC report categories are rounded; percentages may not sum to 100%.

Common financial fraud patterns

Card-not-present	Stolen card details used in online purchases.
Account takeover	Criminal gains access using stolen credentials or session tokens.
Synthetic identity	Real and fabricated identity details are combined.
Payment / APP scam	Victim is manipulated into authorising a transfer.
Insider / merchant abuse	Trusted access or refund flows are exploited.

Proposed end-to-end system architecture



Security & governance across every layer: encryption • access control • audit logs • privacy • model monitoring

Deployment pattern: streaming for transaction decisions; scheduled batch jobs for financial forecasts and model refresh.

Data is the foundation: signals become features

DATA DOMAIN	RAW SIGNAL EXAMPLES	ENGINEERED FEATURES
Transaction	Amount, currency, merchant category, timestamp	Spending deviation; amount percentile; velocity
Customer	Tenure, normal spend, usual locations	Customer baseline; behavioural change
Device & session	Device ID, IP, browser, SIM / device changes	New-device flag; device reputation; IP risk
Network / graph	Shared devices, accounts, beneficiaries	Connected-account count; suspicious clusters
Business / market	Revenue, invoices, cash flows, seasonality	Lagged values; rolling averages; seasonal index

QUALITY CHECK

Deduplicate, validate ranges, handle missing values.

TIME SAFETY

Only use information available at decision time.

PRIVACY

Minimise sensitive data; protect identifiers.

Two AI tracks: forecast the future, score the present

FINANCIAL PREDICTION

Targets: revenue, expenses, cash balance, transaction volume.

Models: linear regression, ARIMA / Prophet-style time series, gradient boosting.

Output: forecast + confidence interval.

FRAUD DETECTION

Target: probability that a transaction is fraudulent.

Models: logistic regression, random forest, gradient boosting, neural networks, anomaly detection.

Output: risk score + reason codes.

Model selection guide

Need	Useful starting point	Why
Explainability	Logistic regression / shallow tree	Easy to audit and communicate
Tabular accuracy	Gradient boosting	Strong on mixed structured features
Sequence patterns	RNN / Transformer	Can learn temporal behaviour
Rare / novel events	Anomaly detection	Flags behaviour unlike known baseline

Real-time transaction decision: from swipe to action

1

Ingest

Payment event arrives with amount, merchant, device and time.

2

Enrich

Fetch customer baseline, device reputation and recent velocity.

3

Score

Rules + ML estimate fraud probability in milliseconds.

4

Decide

Low risk: approve • medium: step-up verification • high: hold / decline.

5

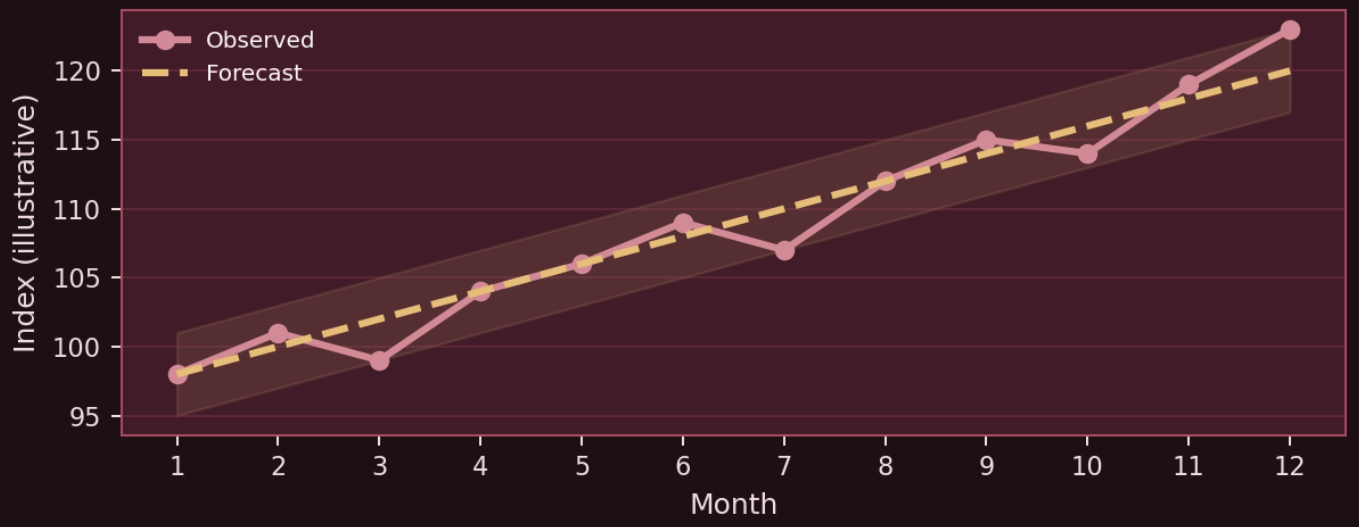
Learn

Customer / investigator outcome becomes a training label.

Important: risk thresholds are tuned to business tolerance; high-risk cases can be routed to human review.

Financial prediction: turning history into a plan

Example: monthly transaction-volume forecast



Typical forecasting workflow

- Collect daily / weekly revenue, expense and balance data.
- Create lag, rolling-average and calendar features.
- Train using chronological time-based splits.
- Compare forecast with actuals; track error over time.
- Publish base, optimistic and downside scenarios.

Chart data is illustrative, not a bank forecast.

Real-world example: Yapı Kredi’s fraud programme

98.7%
Reported reduction in fraud losses over seven years.

40M
Transactions analysed each day.

500
Potential fraud cases detected daily (reported).

2×
Transaction volume doubled while maintaining low fraud ratio.

What changed in the approach?

- Moved beyond static rules to advanced fraud analytics.
- Added custom machine-learning scores to fraud decisioning.
- Scaled monitoring to high-volume, real-time payment activity.
- Used ongoing model performance and operational feedback.

LESSON FOR THIS PROJECT

A measurable fraud programme combines data, ML scoring, operational decisions and continuous monitoring—not just a clever algorithm.

Source: FICO announcement, 30 Jan 2025. Vendor-reported case-study outcomes; not an independent causal evaluation.

Walk-through: how the system catches a suspicious payment

A customer’s normal pattern

SIGNAL	USUAL PATTERN	NEW EVENT
Amount	₹500–₹4,000	₹78,000
Location	Tirupati / Chennai	New overseas IP
Device	Known phone	Unrecognised device
Velocity	1–3 payments / hour	8 attempts / 2 minutes

Decision path (illustrative scores)

Amount deviation — High

Device novelty — High

Velocity spike — High

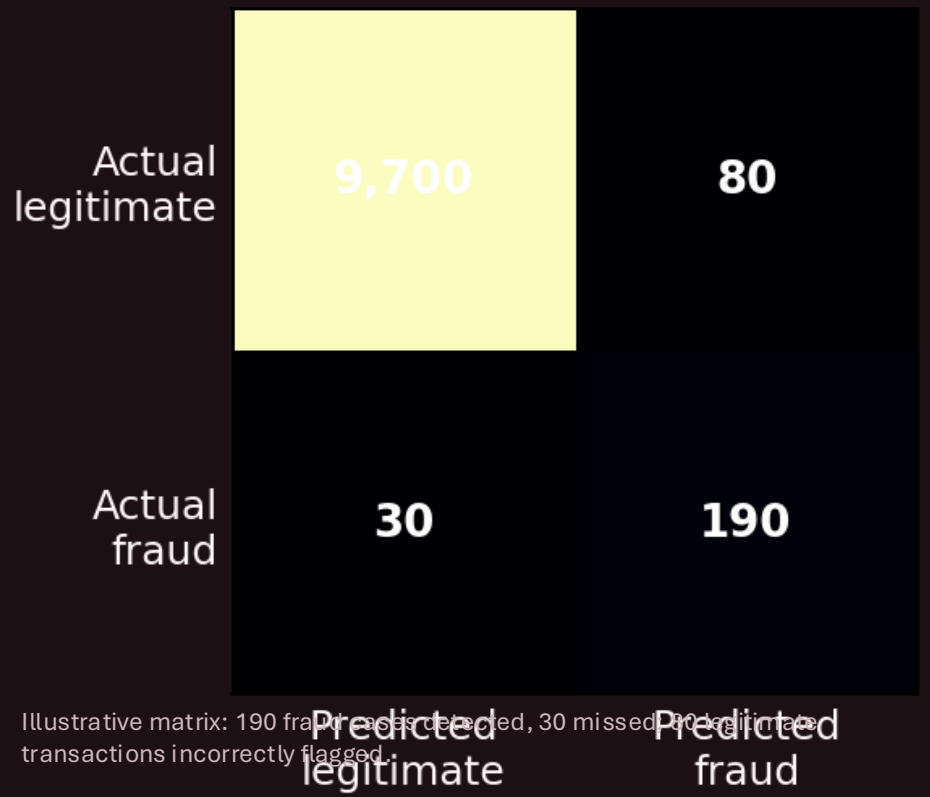
Combined model score — 0.97 / 1.00

ACTION: temporarily hold payment → send secure verification → release only after customer confirms.

Scenario is fictional for teaching. It demonstrates explainable signals and a safer step-up flow.

Measure what matters: accuracy alone is not enough

Illustrative confusion matrix



Illustrative matrix: 190 fraud cases detected, 30 missed, 80 legitimate transactions incorrectly flagged.

Key metrics for fraud models

Precision	Of flagged transactions, how many are truly fraud?
Recall	Of all fraud cases, how many did we catch?
F1-score	Balance between precision and recall.
PR-AUC	Useful when fraud is a very small minority class.
False-positive rate	How often genuine transactions are blocked.

Evaluate by time period, customer segment and fraud type.

Critical challenges—and how a robust system fixes them

CHALLENGE	WHY IT HURTS	PRACTICAL FIX
Class imbalance	Fraud is rare; accuracy can look high while fraud is missed.	Use PR-AUC, class weights, careful resampling and threshold tuning.
Concept drift	Criminal tactics and customer behaviour change.	Monitor drift, refresh labels, retrain and champion/challenger test.
False declines	Legitimate customers are disrupted.	Use risk tiers, step-up verification and cost-sensitive thresholds.
Data leakage	Future information inflates offline results.	Chronological splits; point-in-time feature availability.
Privacy & security	Financial data is sensitive and valuable.	Encryption, least privilege, tokenisation, retention limits and audit logs.
Explainability	Teams need to understand why a payment was flagged.	Reason codes, feature attribution, analyst review and appeal paths.

Operational success = fraud prevented + genuine payments approved + customers treated fairly.

Conclusion: intelligent finance must stay accountable

PREDICT

Use time-aware forecasting to plan cash, revenue and risk.

PROTECT

Combine rules, ML scores and verification to reduce fraud.

IMPROVE

Monitor drift, explain decisions and learn from outcomes.

Selected references

1. US FTC, Consumer Sentinel Network Data Book 2024 (2025). ftc.gov/reports/consumer-sentinel-network-data-book-2024

2. US FTC, “New FTC Data Show a Big Jump in Reported Losses to Fraud to $12.5 Billion in 2024” (Mar 2025). ftc.gov

3. FICO, “Yapı Kredi Cuts Fraud by 98% in Seven Years Using FICO AI Technology” (30 Jan 2025). investors.fico.com

4. Hafez et al., “A systematic review of AI-enhanced techniques in credit card fraud detection,” Journal of Big Data (2025). doi:10.1186/s40537-024-01048-8

5. Cherif et al., “Credit card fraud detection in the era of disruptive technologies: A systematic review,” JKSUCIS (2023). doi:10.1016/j.jksuci.2022.11.008

THANK YOU • Questions?